CASE METHOD 2

PRAKTIKUM ALGORITMA DAN STRUKTUR DATA

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Kode

- Pesanan

pesanan.java > Java > pesanan

```java
public class pesanan {
    int kodePesanan;
    String namaPesanan;
    int harga;

    public pesanan(int kodePesanan, String namaPesanan, int harga) {
        this.kodePesanan = kodePesanan;
        this.namaPesanan = namaPesanan;
        this.harga = harga;
    }
}
```

- Pembeli

pembeli.java > Language Support for Java(TM) by Red Hat > pembeli > pembeli(String, String)

```java
public class pembeli {
    String namaPembeli;
    String noHp;

    public pembeli(String namaPembeli, String noHp) {
        this.namaPembeli = namaPembeli;
        this.noHp = noHp;
    }
}
```

- nodePesanan

nodePesanan.java > Java > nodePesanan

```java
public class nodePesanan {
    int nomorAntrean;
    pembeli pembeli;
    pesanan data;
    nodePesanan prev, next;

    public nodePesanan(nodePesanan prev,
                        int nomorAntrean,
                        pembeli pembeli,
                        pesanan data,
                        nodePesanan next) {
        this.prev = prev;
        this.nomorAntrean = nomorAntrean;
        this.pembeli = pembeli;
        this.data = data;
        this.next = next;
    }
}
```

- nodeAntrian

nodeAntrean.java > Java > nodeAntrean > nodeAntrean(nodeAntrean prev, int nomorAnt

```java
public class nodeAntrean {
    int nomorAntrean;
    pembeli data;
    nodeAntrean prev, next;

    public nodeAntrean(nodeAntrean prev, int nomorAntrean,
                       pembeli data, nodeAntrean next) {
        this.prev = prev;
        this.nomorAntrean = nomorAntrean;
        this.data = data;
        this.next = next;
    }
}
```

- DLLPesanan

DoubleLinkedListPesanan.java > Language Support for Java(TM) by Red Hat > DoubleLinkedListPesan

```java
public class DoubleLinkedListPesanan {
    nodePesanan head;
    nodePesanan tail;

    public boolean isEmpty() {
        return head == null;
    }

    public void addLast(int nomorAntrean, pembeli pembeli, pesanan
        nodePesanan newNode = new nodePesanan(
                tail,
                nomorAntrean,
                pembeli,
                pesanan,
                next: null
        );

        if (isEmpty()) {
            head = tail = newNode;
        } else {
            tail.next = newNode;
            newNode.prev = tail;
            tail = newNode;
        }

        System.out.println(x: "Pesanan berhasil disimpan.");
    }

    public void sortByNamaPesanan() {
        if (isEmpty() || head.next == null) {
            return;
        }

        boolean swapped;

        do {
            swapped = false;
            nodePesanan current = head;

            while (current.next != null) {
                if (current.data.namaPesanan.compareToIgnoreCase(
                        current.next.data.namaPesanan) > 0) {

                            int tempNomor = current.nomorAntrean;
                    pembeli tempPembeli = current.pembeli;
                    pesanan tempPesanan = current.data;

                    current.nomorAntrean = current.next.nomorAntre
                    current.pembeli = current.next.pembeli;
                    current.data = current.next.data;

                    current.next.nomorAntrean = tempNomor;
                    current.next.pembeli = tempPembeli;
                    current.next.data = tempPesanan;

                    swapped = true;
                }
                current = current.next;
            }
        } while (swapped);
    }

    public void printLaporan() {
        if (isEmpty()) {
            System.out.println(x: "Belum ada pesanan.");
            return;
        }

        sortByNamaPesanan();

        System.out.println(x: "\n=== LAPORAN PESANAN (URUT NAMA PE
        System.out.println(x: "Kode\tMenu\t\tHarga\tPembeli");
```

```java
        int total = 0;
        nodePesanan current = head;

        while (current != null) {
            System.out.println(
                    current.data.kodePesanan + "\t" +
                    current.data.namaPesanan + "\t\t" +
                    current.data.harga + "\t" +
                    current.pembeli.namaPembeli
            );

            total += current.data.harga;
            current = current.next;
        }

        System.out.println(x: "----------------------------------------
        System.out.println("Total Pendapatan: Rp " + total);
    }

}
```

- DLLAntrian

```java
public class DoubleLinkedListAntrean {
    nodeAntrean head;
    nodeAntrean tail;
    int nomorOtomatis = 1;

    public boolean isEmpty() {
        return head == null;
    }
```

DoubleLinkedListAntrean.java > Java > DoubleLinkedListAntrean > print()

```java
public class DoubleLinkedListAntrean {
    ...

    public void addLast(pembeli pembeli) {
        nodeAntrean newNode = new nodeAntrean(
                tail,
                nomorOtomatis++,
                pembeli,
                next: null
        );

        if (isEmpty()) {
            head = tail = newNode;
        } else {
            tail.next = newNode;
            newNode.prev = tail;
            tail = newNode;
        }
        System.out.println(x: "Data berhasil ditambahkan ke antrea
    }

    public void print() {
        if (isEmpty()) {
            System.out.println(x: "Antrean masih kosong.");
            return;
        }

        System.out.println(x: "\n--- DAFTAR ANTREAN ---");
        System.out.println(x: "No\tNama\t\tNo HP");

        nodeAntrean current = head;
        while (current != null) {
            System.out.println(
                    current.nomorAntrean + "\t" +
                    current.data.namaPembeli + "\t\t" +
                    current.data.noHp
            );
            current = current.next;
        }
    }

    public nodeAntrean removeFirst() {
        if (isEmpty()) {
            System.out.println(x: "Antrean kosong.");
            return null;
        }

        nodeAntrean removed = head;

        if (head == tail) {
            head = tail = null;
        } else {
            head = head.next;
            head.prev = null;
        }

        removed.next = null;
        return removed;
    }

    public int size() {
        int count = 0;
        nodeAntrean current = head;
        while (current != null) {
            count++;
            current = current.next;
        }
        return count;
    }

}
```

- mainKasir

```java
MainKasir.java > Java > MainKasir > main(String[] args)
import java.util.Scanner;

public class MainKasir {
        Run | Debug | Run main | Debug main
        public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        DoubleLinkedListAntrean antrean = new DoubleLinkedListAntr
        DoubleLinkedListPesanan daftarPesanan = new DoubleLinkedLi

        int pilih;

        do {
            System.out.println(x: "\n------------------------------
            System.out.println(x: "    SISTEM ANTREAN RESTO ROYAL D
            System.out.println(x: "------------------------------
            System.out.println(x: "1. Tambah Antrean");
            System.out.println(x: "2. Cetak Antrean");
            System.out.println(x: "3. Hapus Antrean & Input Pesana
            System.out.println(x: "4. Laporan Pesanan");
            System.out.println(x: "0. Keluar");
            System.out.print(x: "Pilih menu: ");
            pilih = sc.nextInt();
            sc.nextLine();

            switch (pilih) {
                case 1:
                    System.out.print(s: "Nama Pembeli : ");
                    String nama = sc.nextLine();

                    System.out.print(s: "No HP        : ");
                    String hp = sc.nextLine();

                    pembeli pembeli = new pembeli(nama, hp);
                    antrean.addLast(pembeli);
                    break;

                case 2:
                    antrean.print();
                    System.out.println("Jumlah antrean: " + antre
                    break;

                case 3:
                    nodeAntrean dipanggil = antrean.removeFirst();

                    if (dipanggil != null) {
                        System.out.println(x: "\nPembeli dipanggil
                        System.out.println("No Antrean : " + dipar
                        System.out.println("Nama       : " + dipar

                        System.out.print(s: "Kode Pesanan : ");
                        int kode = sc.nextInt();
                        sc.nextLine();

                        System.out.print(s: "Nama Menu    : ");
                        String menu = sc.nextLine();

                        System.out.print(s: "Harga        : ");
                        int harga = sc.nextInt();
                        sc.nextLine();

                        pesanan pesanan = new pesanan(kode, menu,

                        daftarPesanan.addLast(
                                dipanggil.nomorAntrean,
                                dipanggil.data,
                                pesanan
                        );
                    }
                    break;

                case 4:
                    daftarPesanan.printLaporan();
                    break;

                case 0:
                    System.out.println(x: "Terima kasih.");
                    break;

                default:
                    System.out.println(x: "Menu tidak valid.");
            }

        } while (pilih != 4);

    }
```

Output

```
====================================
   SISTEM ANTREAN RESTO ROYAL DELISH
====================================
1. Tambah Antrean
2. Cetak Antrean
3. Hapus Antrean & Input Pesanan
4. Laporan Pesanan
0. Keluar
Pilih menu: 1
Nama Pembeli : caca
No HP        : 08132
Data berhasil ditambahkan ke antrean.
```

-

```
====================================
   SISTEM ANTREAN RESTO ROYAL DELISH
====================================
1. Tambah Antrean
2. Cetak Antrean
3. Hapus Antrean & Input Pesanan
4. Laporan Pesanan
0. Keluar
Pilih menu: 2

=== DAFTAR ANTREAN ===
No      Nama            No HP
1       sasa            08123
2       rara            08213
3       caca            08132
Jumlah antrean: 3
```

-

```
====================================
   SISTEM ANTREAN RESTO ROYAL DELISH
====================================
1. Tambah Antrean
2. Cetak Antrean
3. Hapus Antrean & Input Pesanan
4. Laporan Pesanan
0. Keluar
Pilih menu: 3

Pembeli dipanggil:
No Antrean : 1
Nama       : sasa
Kode Pesanan : 123
Nama Menu    : cumi asam manis
Harga        : 25000
Pesanan berhasil disimpan.
```

-

```
====================================
   SISTEM ANTREAN RESTO ROYAL DELISH
====================================
1. Tambah Antrean
2. Cetak Antrean
3. Hapus Antrean & Input Pesanan
4. Laporan Pesanan
0. Keluar
Pilih menu: 4

=== LAPORAN PESANAN (URUT NAMA MENU) ===
Kode    Menu            Harga   Pembeli
123     cumi asam manis         25000   sasa
234     tahu cabe garam         18000   rara
---------------------------------------
Total Pendapatan: Rp 43000
```

-

```
====================================
   SISTEM ANTREAN RESTO ROYAL DELISH
====================================
1. Tambah Antrean
2. Cetak Antrean
3. Hapus Antrean & Input Pesanan
4. Laporan Pesanan
0. Keluar
Pilih menu: 0
Terima kasih.
```

-

```
====================================
   SISTEM ANTREAN RESTO ROYAL DELISH
====================================
1. Tambah Antrean
2. Cetak Antrean
3. Hapus Antrean & Input Pesanan
4. Laporan Pesanan
0. Keluar
Pilih menu: 5
Menu tidak valid.
```

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